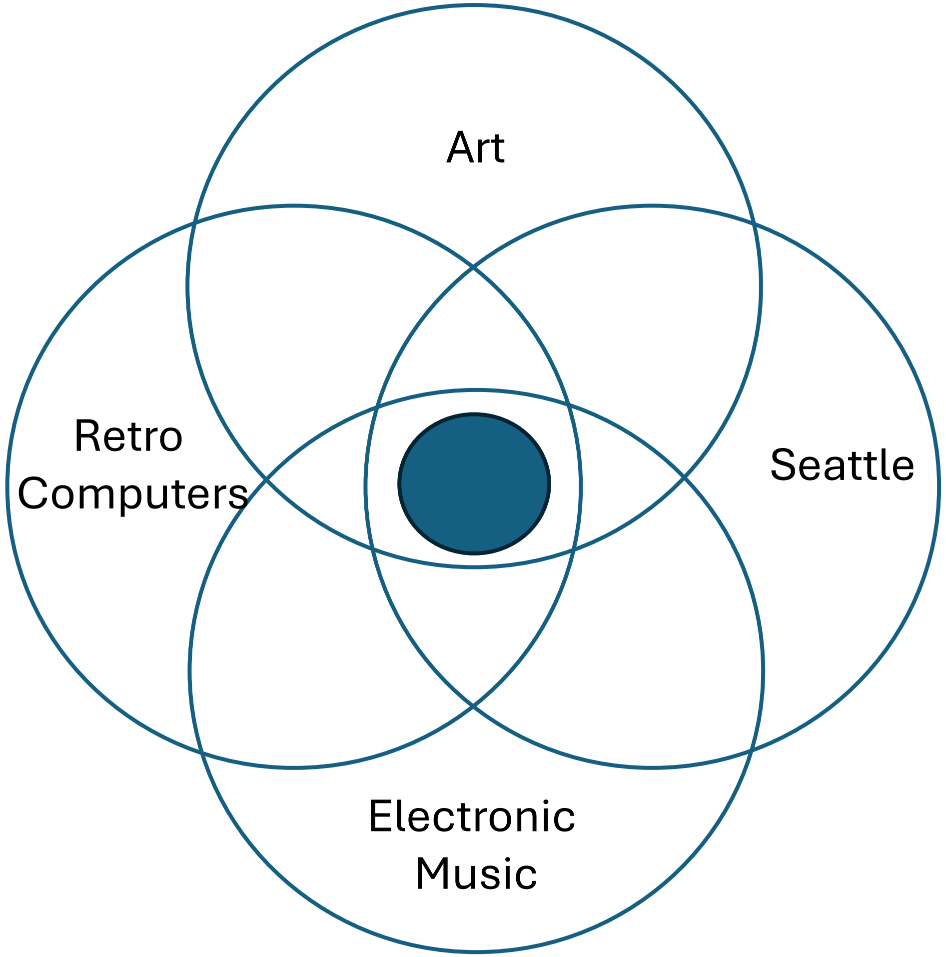
Network IV

SEA-TAC Airport Sound-Light Environment


Art
Retro Computers
Seattle
Electronic Music

Outline

- James Seawright’s Network IV SEA-TAC installation
- Rebooting Network IV

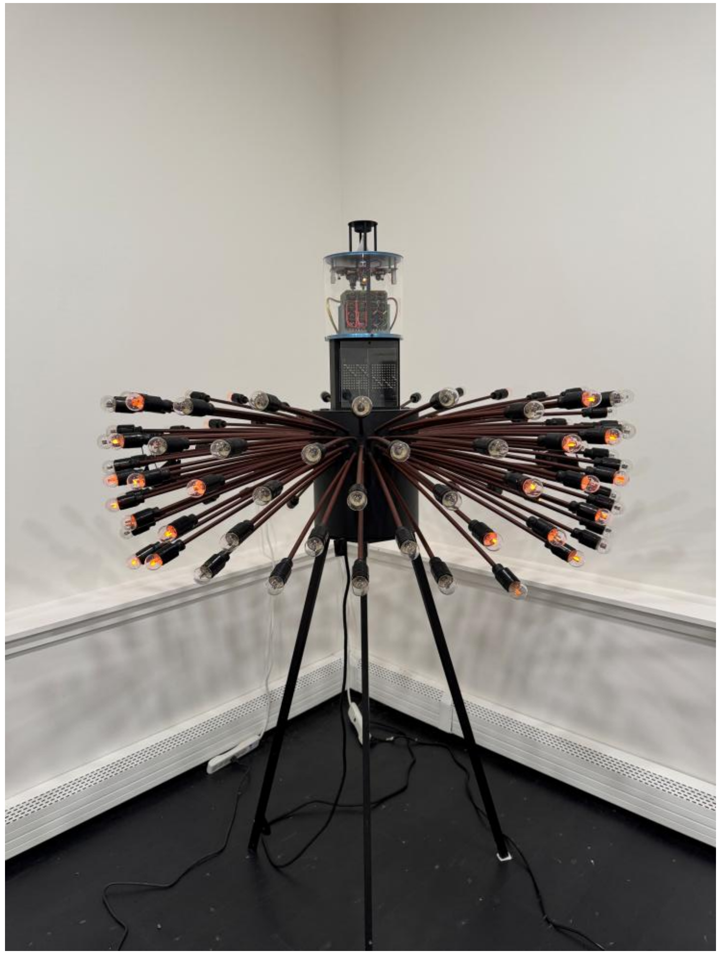
James Seawright 1936 - 2022

- Technological Artist / Kinetic Sculptor
- Director of Visual Arts Princeton University
- Technical Supervisor Columbia-Princeton Electronic Music Center

James Seawright Network Series

NETWORK I
1969

NETWORK II
1969



Sound interactive. Technology: RTL / DTL Logic, Neon Glow Lamps

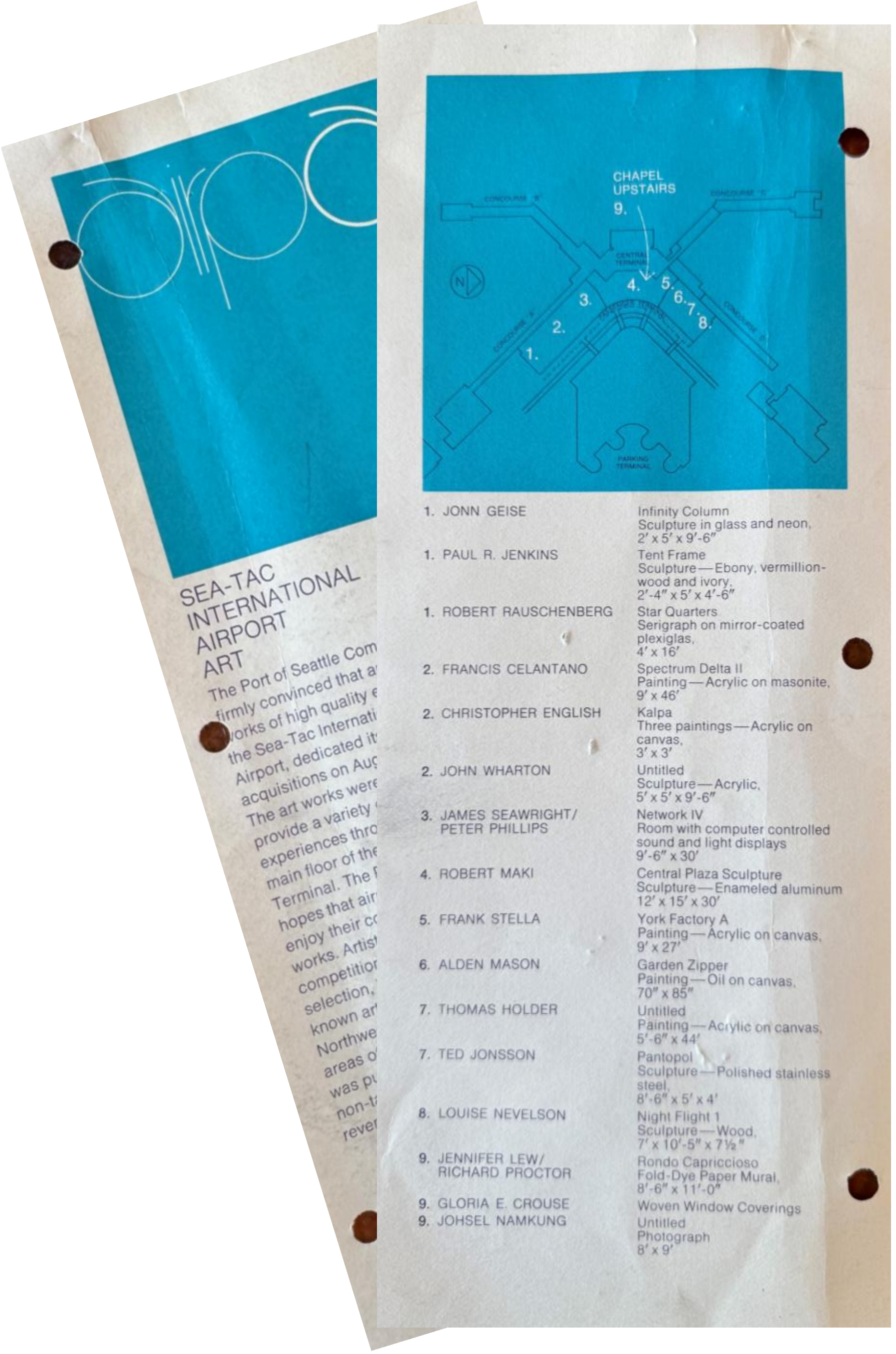
Network III 1971

Installation in the Walker Art Center Minneapolis

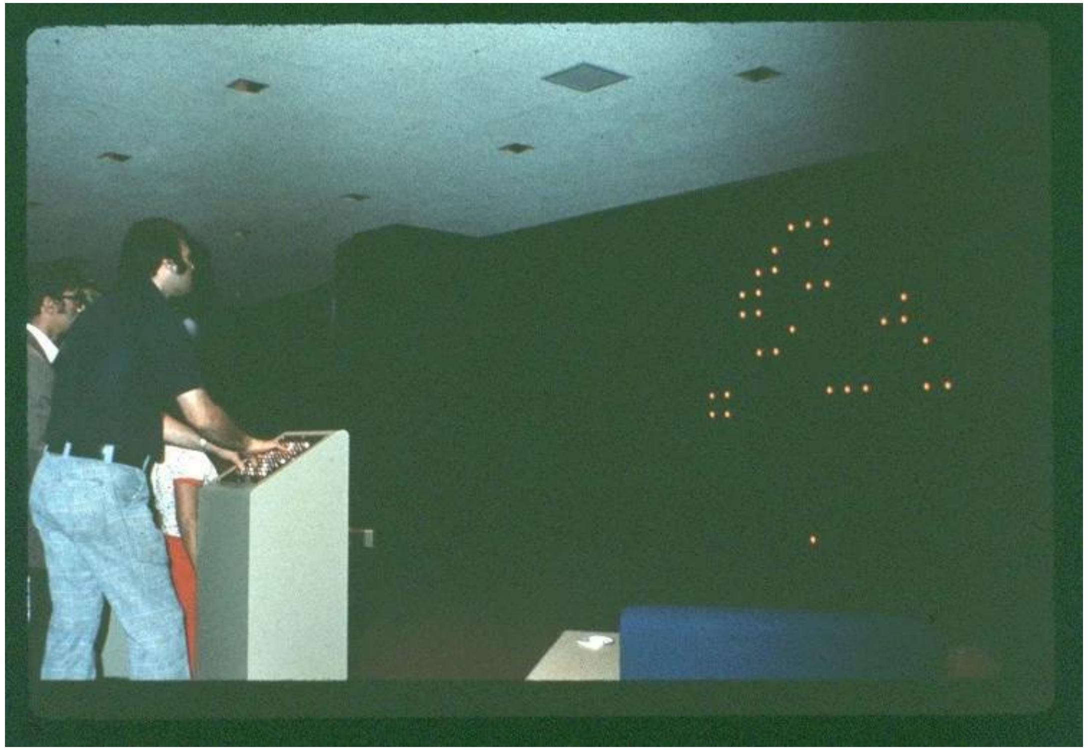
PDP-8

SEA-TAC ART
1973

"The Port of Seattle Commission, firmly convinced that artistic works of high quality enhance the Sea-Tac International Airport, dedicated its art acquisitions on August 1, 1973. ..."

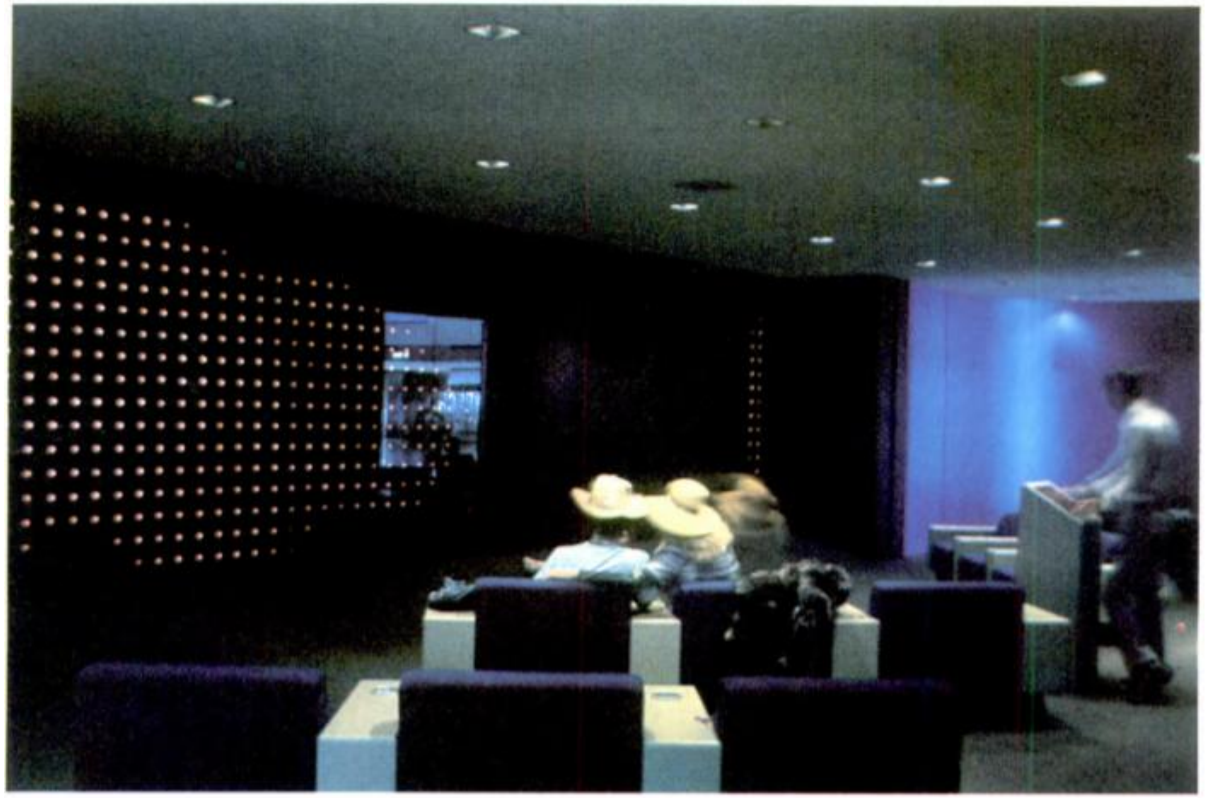


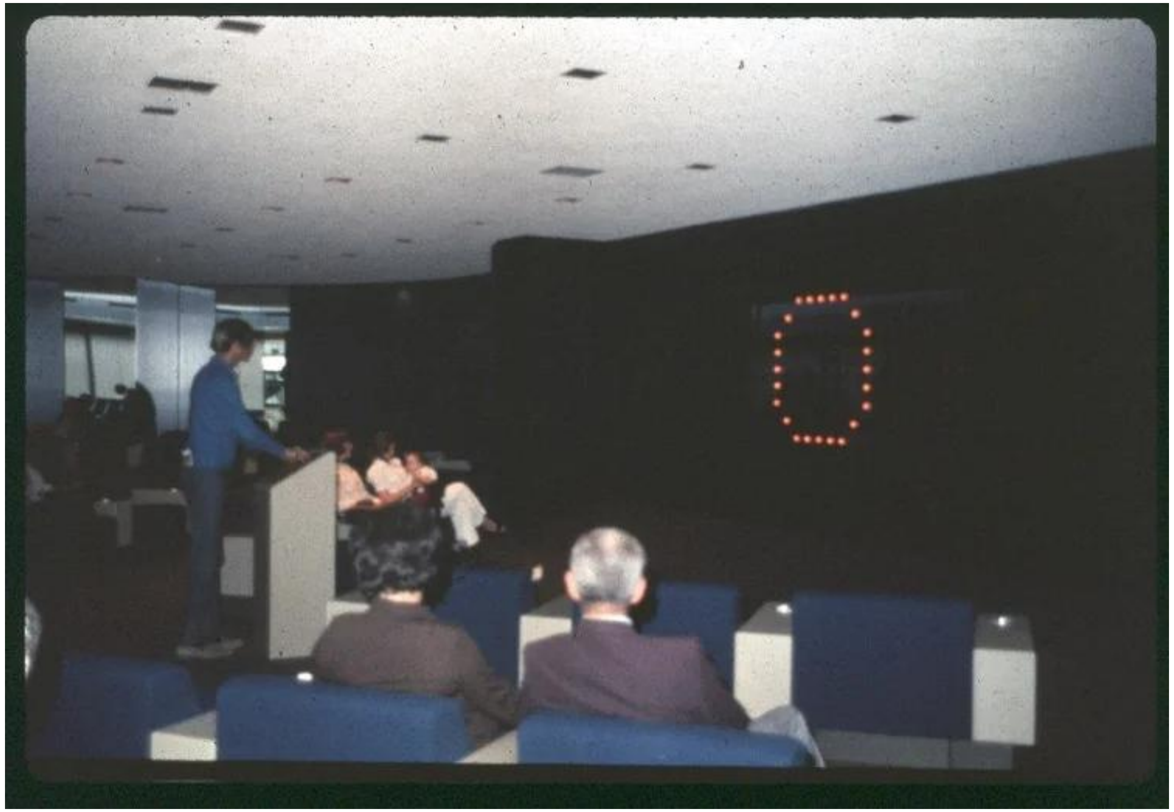
Network IV



Network IV

Electronic environment, James Seawright and Peter Philips, 1974, SEA-TAC Airport.





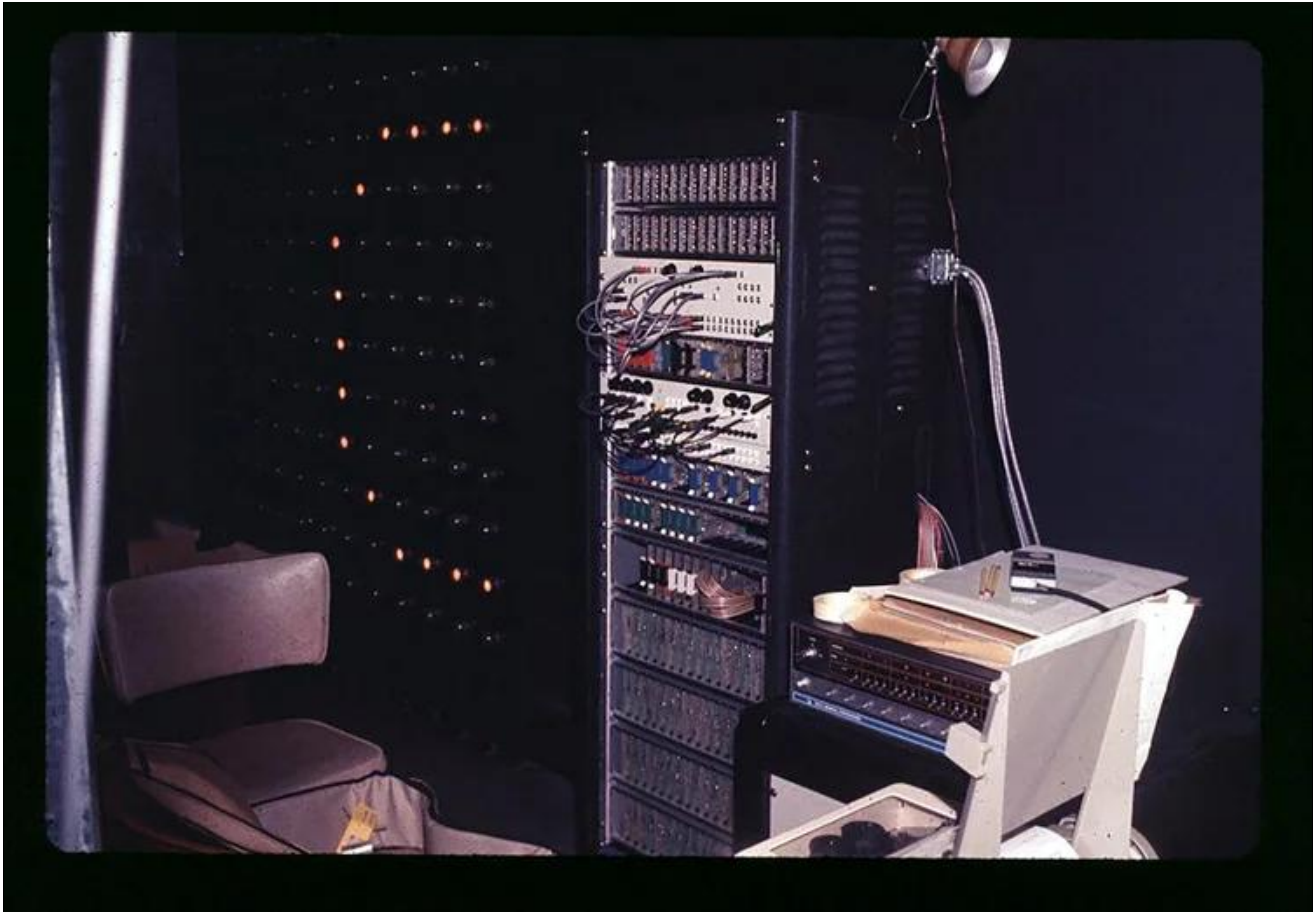
Network IV Behavior

- Idle mode: a blob pattern moves back and forth across the display, seeming to bounce off the top or bottom, and traveling at various rates with a loosely structured sound accompaniment
- Interactive mode: button patterns introduce rapidly changing patterns based on John Conway’s life game appear accompanied by synchronized sound patterns

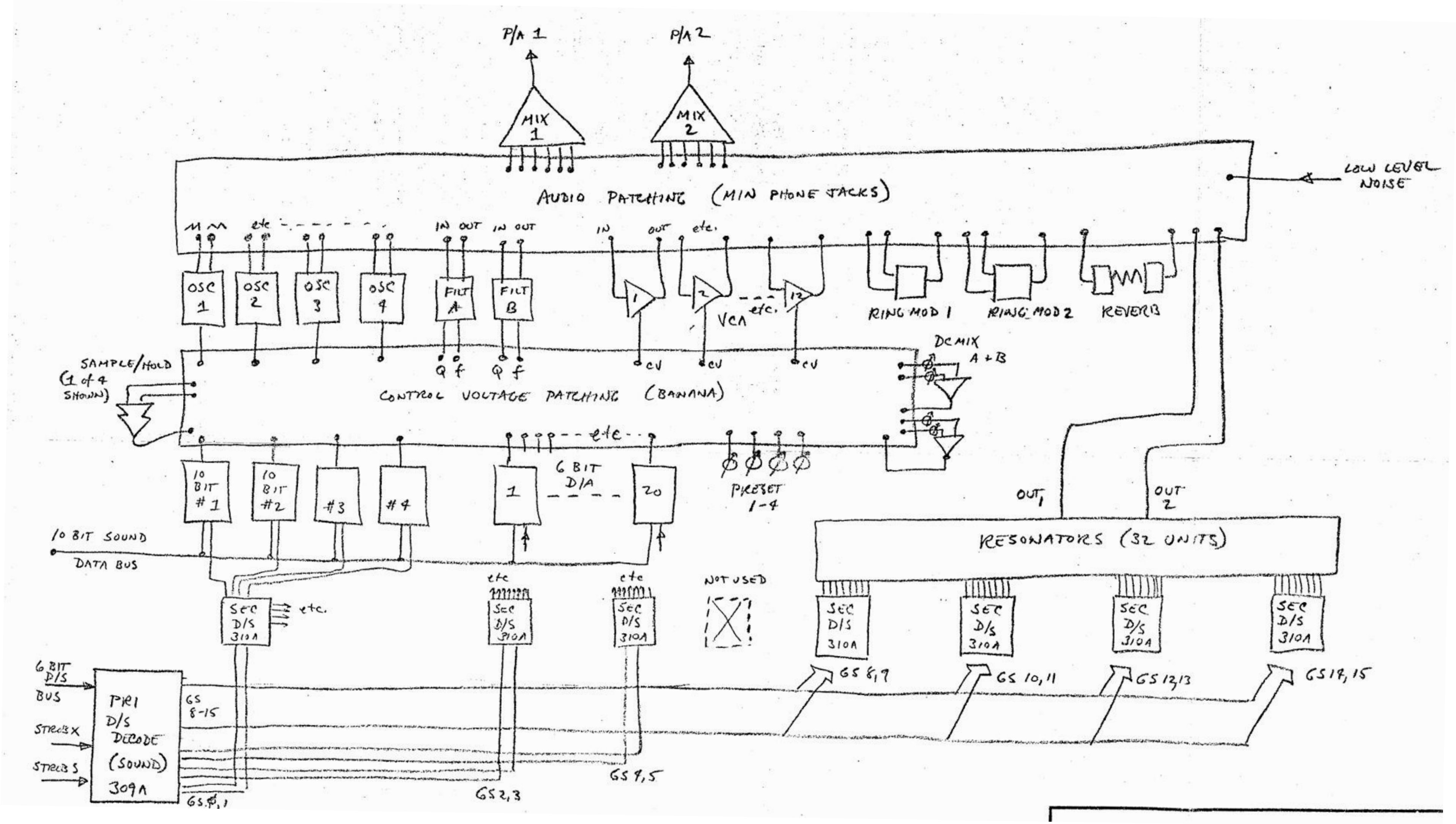
Network IV Under the Hood

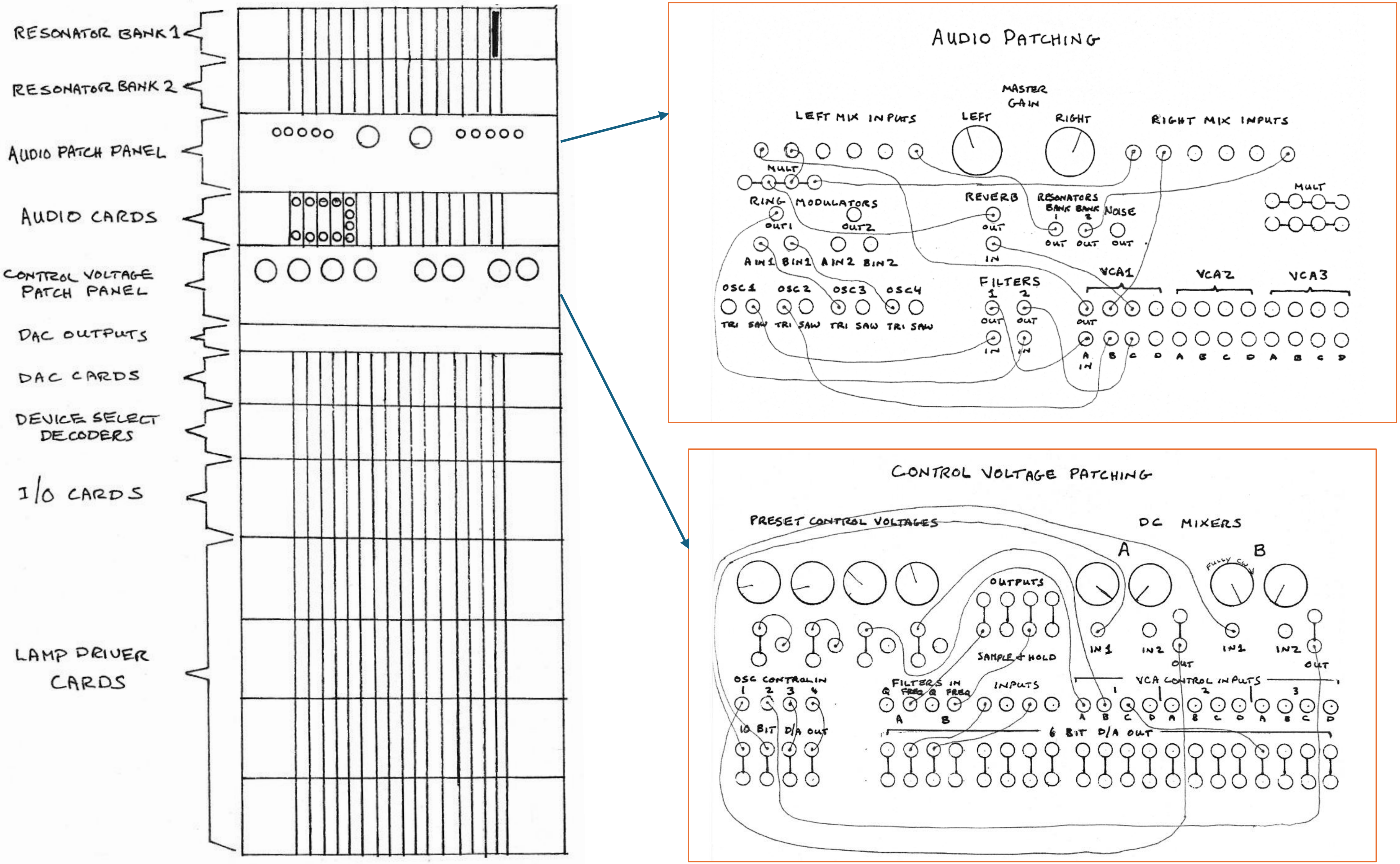
Network IV SEA-TAC Airport Sound-Light Environment

- Display Wall
  - 64 x 16 Neon glow light array
  - 6” centers
  - 8ft high by 32ft long behind plate glass
- Touch plate with 8x8 buttons
- Sound
  - Three Moog-style electronic synthesizer instruments controlled from DACs
  - 32 digitally triggered resonators
- Computer
  - Data General Nova 1210
  - Data General #4040 “General Purpose Interface”



Network IV Sound





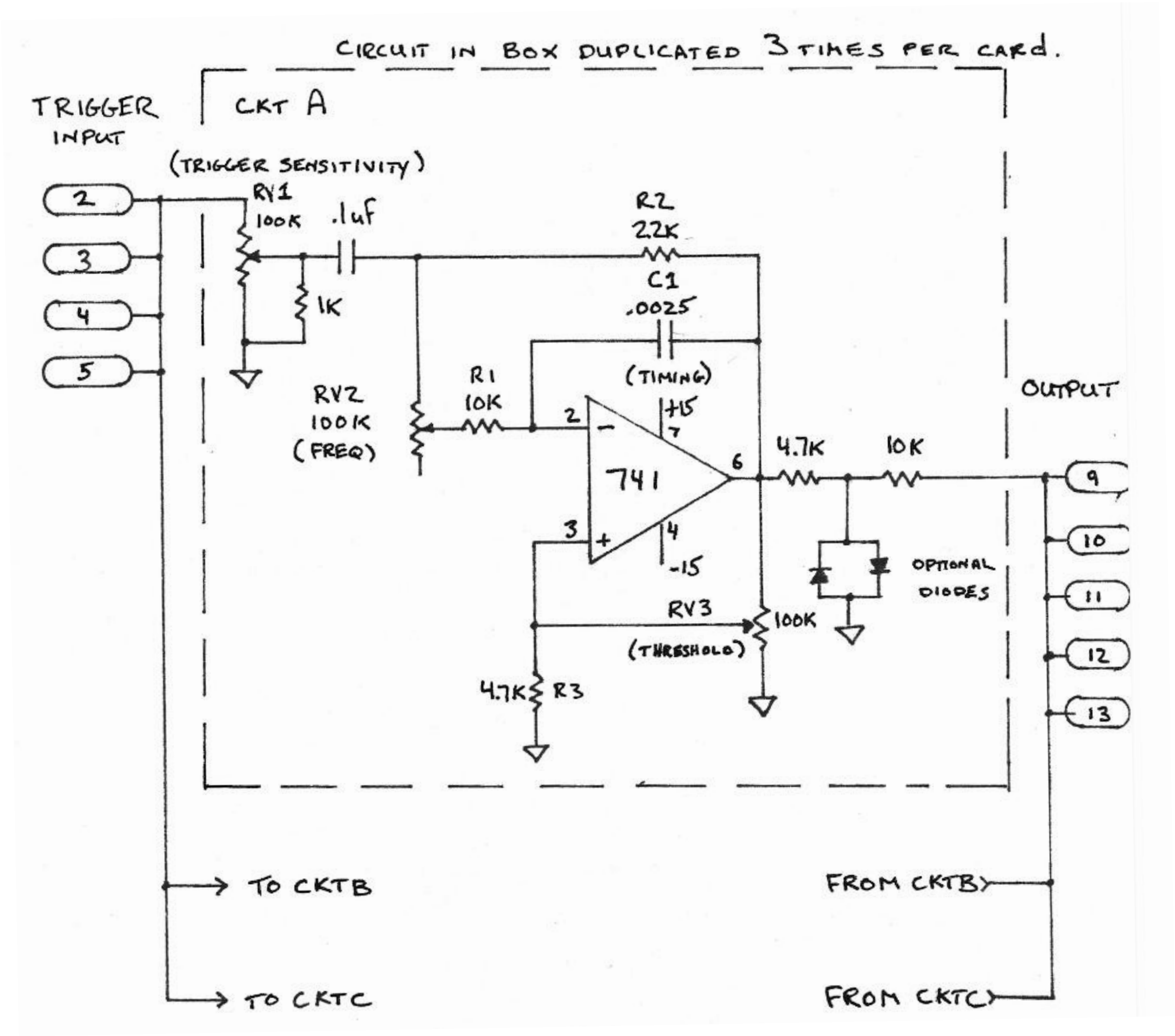
RESONATOR BANK 1
RESONATOR BANK 2
AUDIO PATCH PANEL
AUDIO CARDS
CONTROL VOLTAGE PATCH PANEL
DAC OUTPUTS
DAC CARDS
DEVICE SELECT DECODERS
I/O CARDS
LAMP DRIVER CARDS
AUDIO PATCHING
MASTER GAIN
LEFT MIX INPUTS
LEFT
RIGHT
RIGHT MIX INPUTS
MULT
RING MODULATORS
OUT1
OUT2
A IN 1
B IN 1
A IN 2
B IN 2
REVERB
OUT
IN
RESONATORS
BANK 1
BANK 2
NOISE
OUT
OSC 1
OSC 2
OSC 3
OSC 4
TRI
SAW
FILTERS
1
2
OUT
IN
VCA 1
VCA 2
VCA 3
A
B
C
D
CONTROL VOLTAGE PATCHING
PRESET CONTROL VOLTAGES
DC MIXERS
A
B
FULLY CW
OUTPUTS
SAMPLE & HOLD
INPUTS
IN 1
IN 2
OUT
OSC CONTROL IN
1
2
3
4
FILTERS IN
Q
FREQ
A
B
VCA CONTROL INPUTS
10 BIT D/A OUT
6 BIT D/A OUT

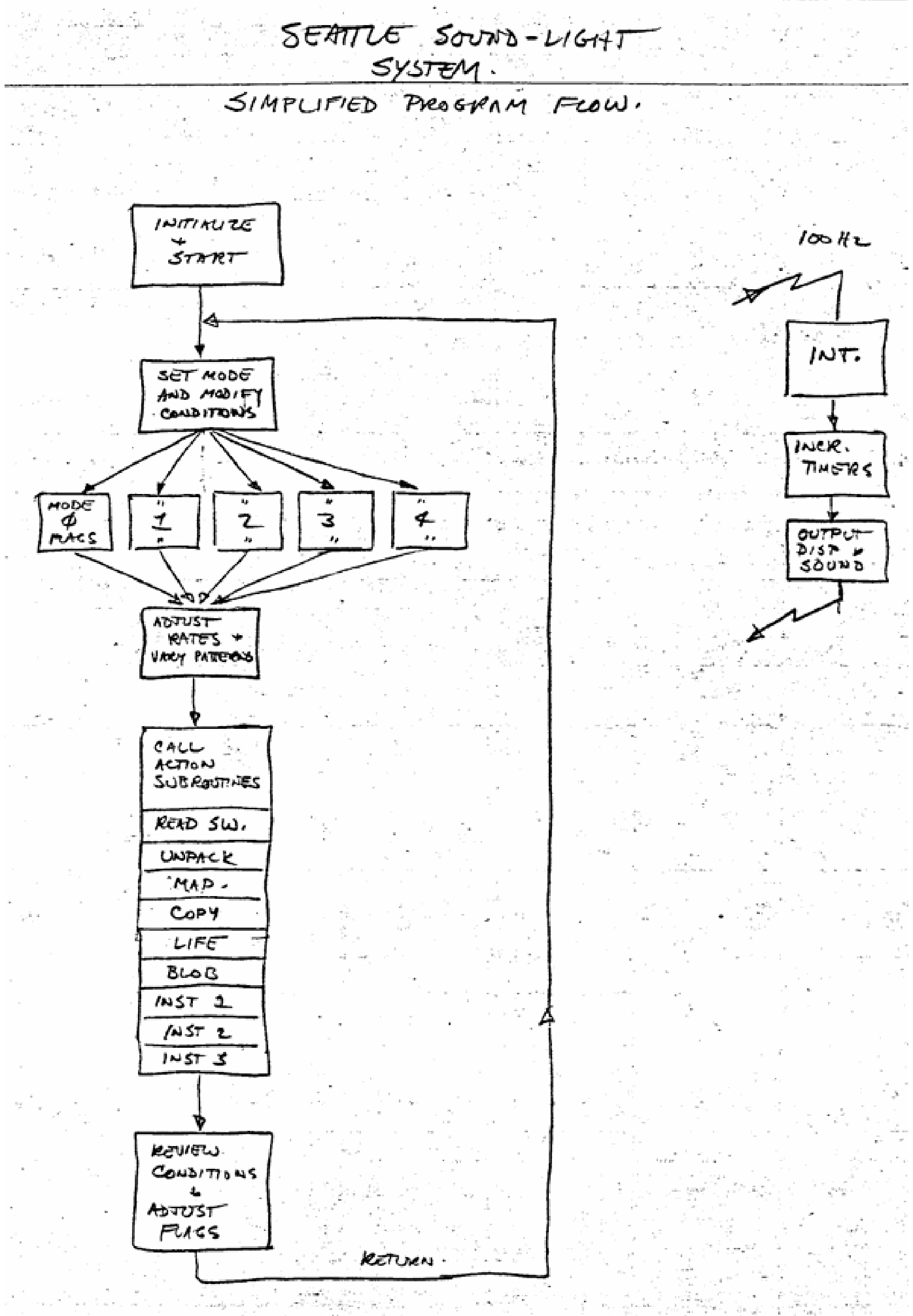
Resonators

32 Resonators

Each one is three 741 circuits producing decaying sinusoidal resonant sound

Triggered by the computer





Program

00400 : Starting address

02000: Interrupt handler

CORE MAP

0 - 3777	MAIN OPERATING PROGRAM
4000 - 4100	PITCH LIST
4400 - 4472	FILTER LIST
5000 - 5100	ENVELOPE LIST
5700 - 5720	MODIFY LIST
6000 - 6674	BLOB SUBROUTINE
7400 - 7577	PUNCH (NOT USUALLY RESIDENT)
10000 - 14777	LIFE AND DISPLAY ASSOCIATED LISTS
15000 - 15400	RESONATER LIST
16000 - 16123	TEST ROUTINES
17600 - 17777	RESERVED FOR LOADERS

Network IV

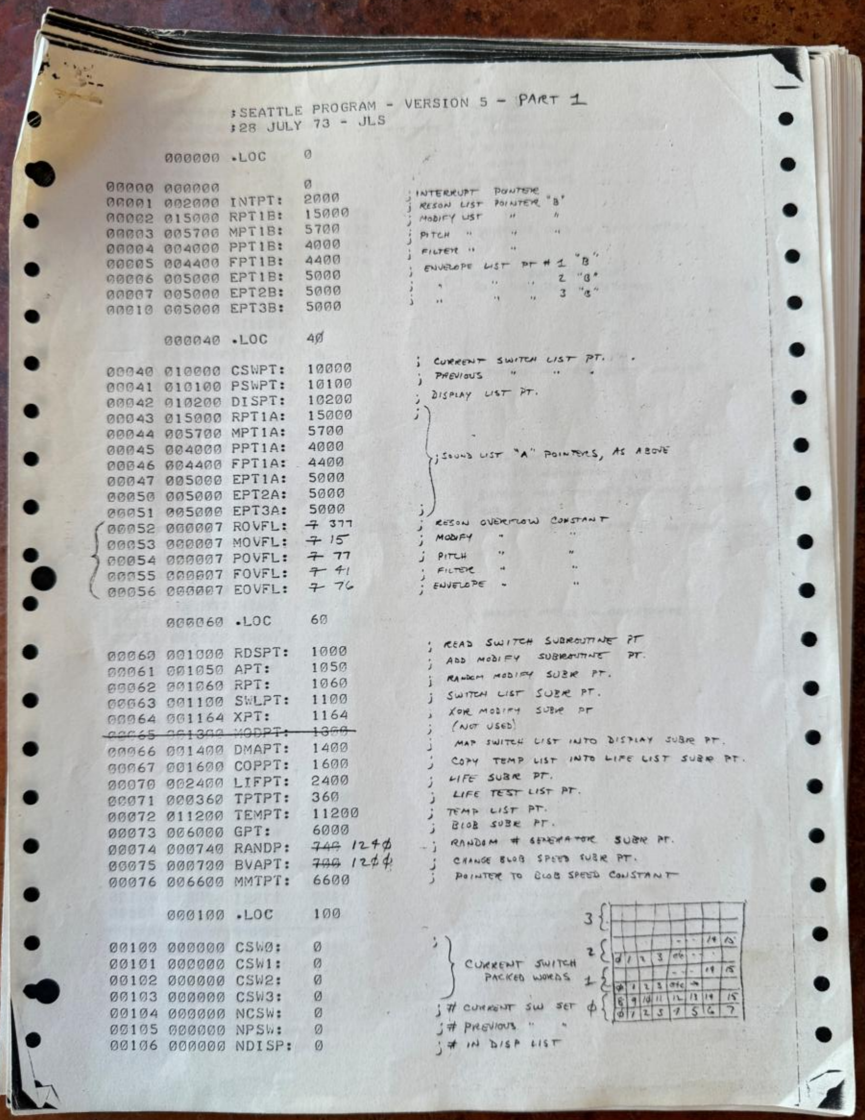
- Network IV was in operation from 1973 – late 90's or early 2000's
- There were overhauls in 1986 and 1988
- The surviving elements
  - neon glow lamps
  - the code
  - documentation

Network IV Rebooted

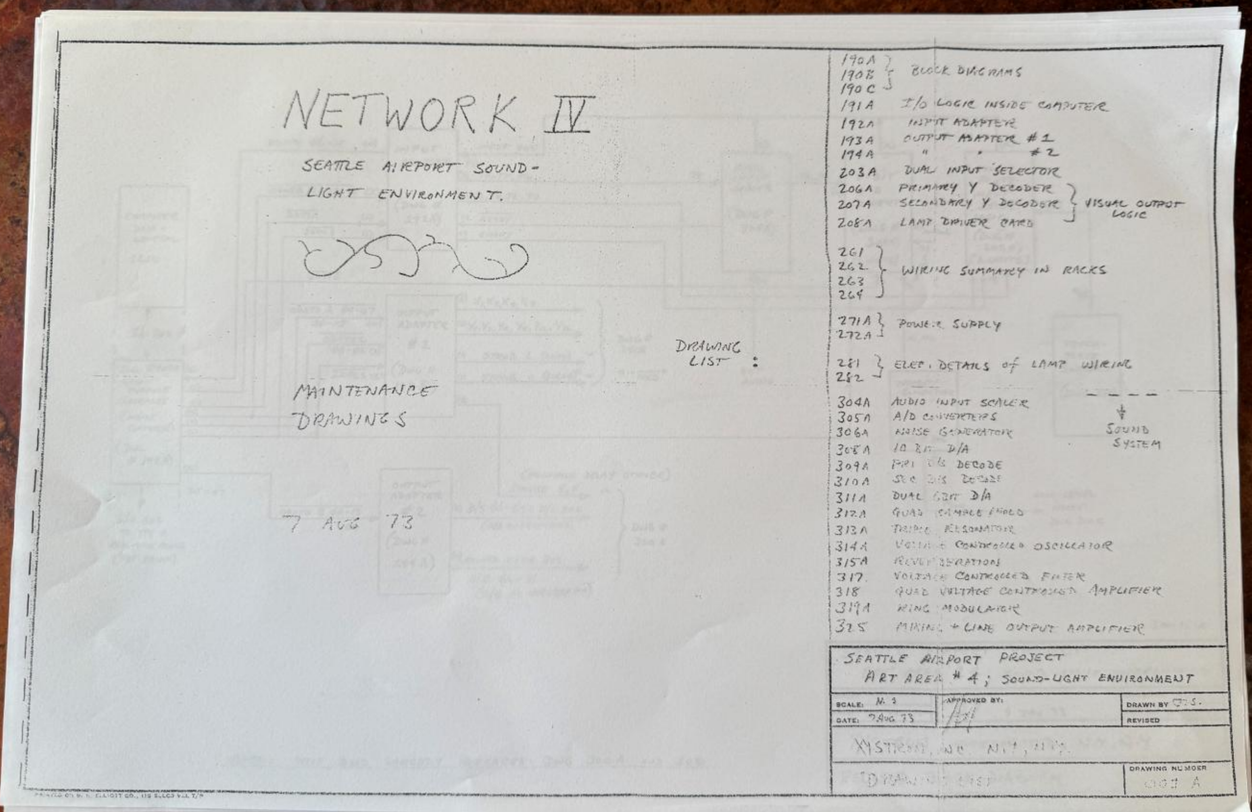
Genesis


SEATTLE PIECE
NETWORK IV
XYSTRON
INCORPORATED
65 EAST SECOND STREET
NEW YORK, N. Y. 10003
FINAL PROG.
FINAL PROGRAM OBJ.
TTY COPY UNVERIFIED

Additional Documentation from Charles Keagle



```
;SEATTLE PROGRAM - VERSION 5 - PART 1
;28 JULY 73 - JLS

      000000 .LOC     0

00000 000000          0
00001 002000 INTPT:   2000      ;INTERRUPT POINTER
00002 015000 RPT1B:   15000     ;RESON LIST POINTER "B"
00003 005700 MPT1B:   5700      ;MODIFY LIST  "    "
00004 004000 PPT1B:   4000      ;PITCH  "     "    "
00005 004400 FPT1B:   4400      ;FILTER "     "
00006 005000 EPT1B:   5000      ;ENVELOPE LIST PT # 1 "B"
00007 005000 EPT2B:   5000      ;  "      "    "   2 "B"
00010 005000 EPT3B:   5000      ;  "      "    "   3 "B"

      000040 .LOC     40

00040 010000 CSWPT:   10000     ;CURRENT SWITCH LIST PT.
00041 010100 PSWPT:   10100     ;PREVIOUS  "     "   "
00042 010200 DISPT:   10200     ;DISPLAY LIST PT.
00043 015000 RPT1A:   15000     ;
00044 005700 MPT1A:   5700
00045 004000 PPT1A:   4000
00046 004400 FPT1A:   4400      ;SOUND LIST "A" POINTERS, AS ABOVE
00047 005000 EPT1A:   5000
00050 005000 EPT2A:   5000
00051 005000 EPT3A:   5000      ;
00052 000007 ROVFL:   7 377     ;RESON OVERFLOW CONSTANT
00053 000007 MOVFL:   7 15      ;MODIFY   "       "
00054 000007 POVFL:   7 77      ;PITCH    "       "
00055 000007 FOVFL:   7 41      ;FILTER   "       "
00056 000007 EOVFL:   7 76      ;ENVELOPE "       "

      000060 .LOC     60

00060 001000 RDSPT:   1000      ;READ SWITCH SUBROUTINE PT
00061 001050 APT:     1050      ;ADD MODIFY SUBROUTINE PT.
00062 001060 RPT:     1060      ;RANDOM MODIFY SUBR PT.
00063 001100 SWLPT:   1100      ;SWITCH LIST SUBR PT.
00064 001164 XPT:     1164      ;XOR MODIFY SUBR PT
~~00065 001300 MODPT: 1300~~    ;(NOT USED)
00066 001400 DMAPT:   1400      ;MAP SWITCH LIST INTO DISPLAY SUBR PT.
00067 001600 COPPT:   1600      ;COPY TEMP LIST INTO LIFE LIST SUBR PT.
00070 002400 LIFPT:   2400      ;LIFE SUBR PT.
00071 000360 TPTPT:   360       ;LIFE TEST LIST PT.
00072 011200 TEMPT:   11200     ;TEMP LIST PT.
00073 006000 GPT:     6000      ;BLOB SUBR PT.
00074 000740 RANDP:   ~~740~~ 1240  ;RANDOM # GENERATOR SUBR PT.
00075 000700 BVAPT:   ~~700~~ 1200  ;CHANGE BLOB SPEED SUBR PT.
00076 006600 MMTPT:   6600      ;POINTER TO BLOB SPEED CONSTANT

      000100 .LOC     100

00100 000000 CSW0:    0         ;
00101 000000 CSW1:    0         ; CURRENT SWITCH
00102 000000 CSW2:    0         ;  PACKED WORDS
00103 000000 CSW3:    0
00104 000000 NCSW:    0         ;# CURRENT SW SET
00105 000000 NPSW:    0         ;# PREVIOUS "   "
00106 000000 NDISP:   0         ;# IN DISP LIST
```



NETWORK IV

SEATTLE AIRPORT SOUND-LIGHT ENVIRONMENT.

MAINTENANCE DRAWINGS

7 AUG 73

DRAWING LIST :

190A, 190B, 190C — BLOCK DIAGRAMS
191A I/O LOGIC INSIDE COMPUTER
192A INPUT ADAPTER
193A OUTPUT ADAPTER #1
194A " " #2
203A DUAL INPUT SELECTOR
206A PRIMARY Y DECODER
207A SECONDARY Y DECODER
208A LAMP DRIVER CARD
(206A–208A) VISUAL OUTPUT LOGIC
261, 262, 263, 264 — WIRING SUMMARY IN RACKS
271A, 272A — POWER SUPPLY
281, 282 — ELEC. DETAILS of LAMP WIRING
304A AUDIO INPUT SCALER
305A A/D CONVERTERS
306A NOISE GENERATOR
308A 10 BIT D/A
309A PRI [illegible] DECODE
310A SEC [illegible] DECODE
311A DUAL [illegible] D/A
312A QUAD SAMPLE HOLD
313A TRIPLE RESONATOR
314A VOLTAGE CONTROLLED OSCILLATOR
315A REVERBERATION
317 VOLTAGE CONTROLLED FILTER
318 QUAD VOLTAGE CONTROLLED AMPLIFIER
319A RING MODULATOR
325 MIXING + LINE OUTPUT AMPLIFIER
(304A–325) SOUND SYSTEM

SEATTLE AIRPORT PROJECT
ART AREA # 4; SOUND-LIGHT ENVIRONMENT

SCALE: N. S.	APPROVED BY:	DRAWN BY [illegible]
DATE: 7 AUG 73		REVISED

XYSTRON, INC. N.Y., N.Y.

DRAWING LIST — DRAWING NUMBER [illegible] A

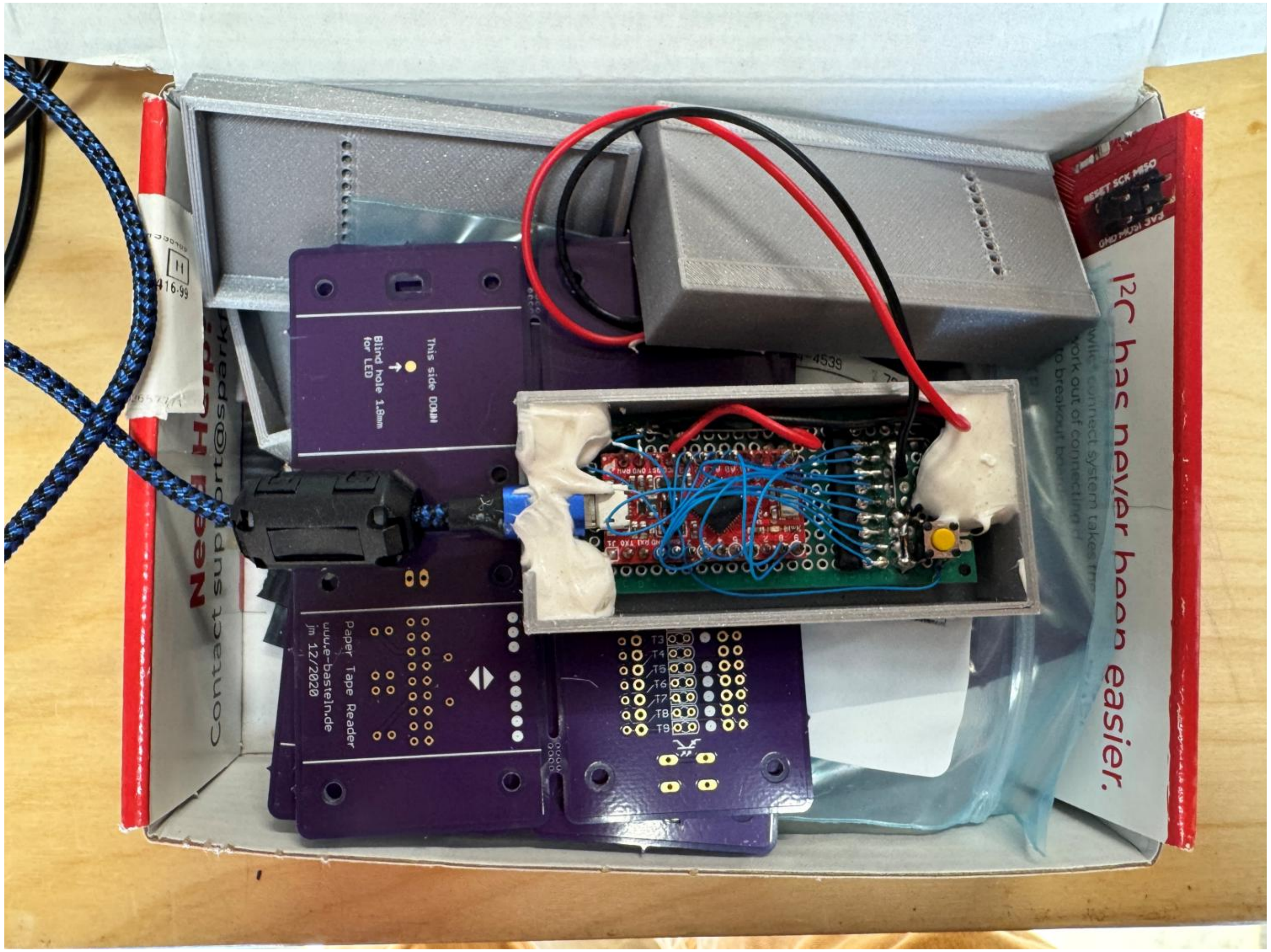
Project threads

- How to convert the paper tape ?
- Understand SIMH NOVA
- IO mapping and timer interrupt ?
- Sound modeling
- Display and input

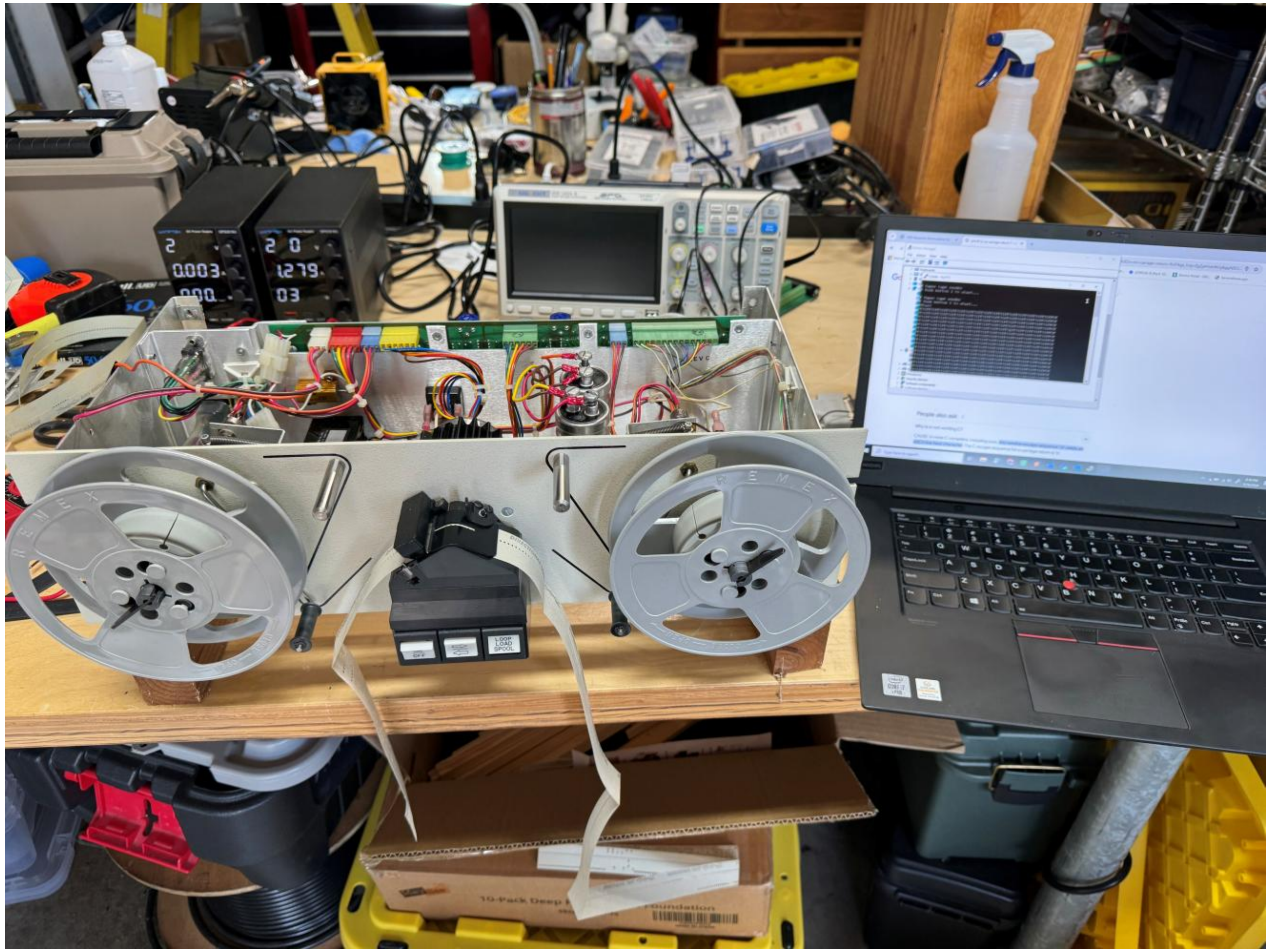
Paper Tape Conversion Saga

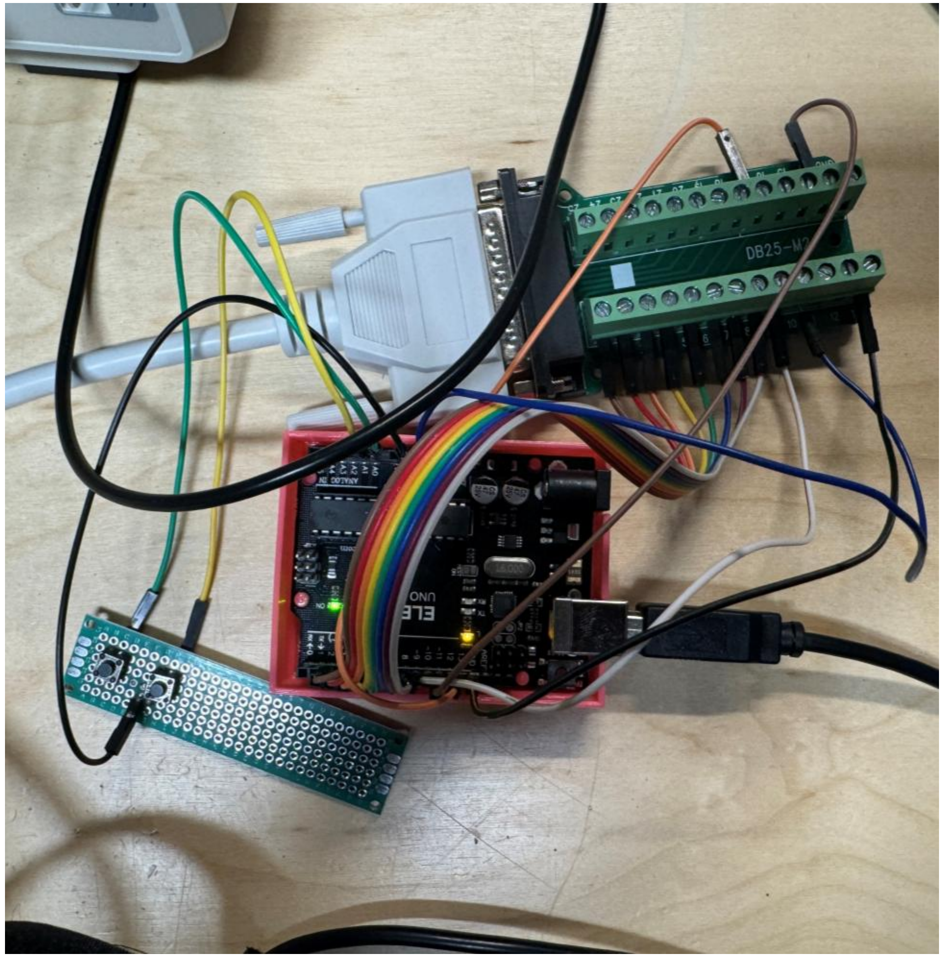
- Built and tried a couple open-source DIY “hand pulled” AVR based readers
  - Not reliable enough ☹
- Purchased REMEX RRS7155BA1 reader on ebay
  - Inspired by Robert Baruch https://www.youtube.com/watch?v=dq2EuooRrVc
  - REMEX reader had power supply issue ☹
- Backup plan – started hand entering from assembled listing (~30% completed)
- REMEX working ☺ Bypass REMEX power supply with bench supplies, fixed bad filter cap later
- Adruino UNO to control REMEX and output data as HEX to laptop
  - Video: https://www.youtube.com/shorts/PvraiOrcCRk
- Understand Nova paper tape format
- Python script to convert tape data in Nova format to SIMH deposit commands
  - Checksums all match! ☺
- Comparing hand entered code from listing to tape data matched ☺
- Data files, conversion scripts and Arduino code is on github

Paper tape first attempts



Tape Reader Setup: REMEX RRS7155BA1 + Arduino -> Ascii Hex -> Laptop Serial Port





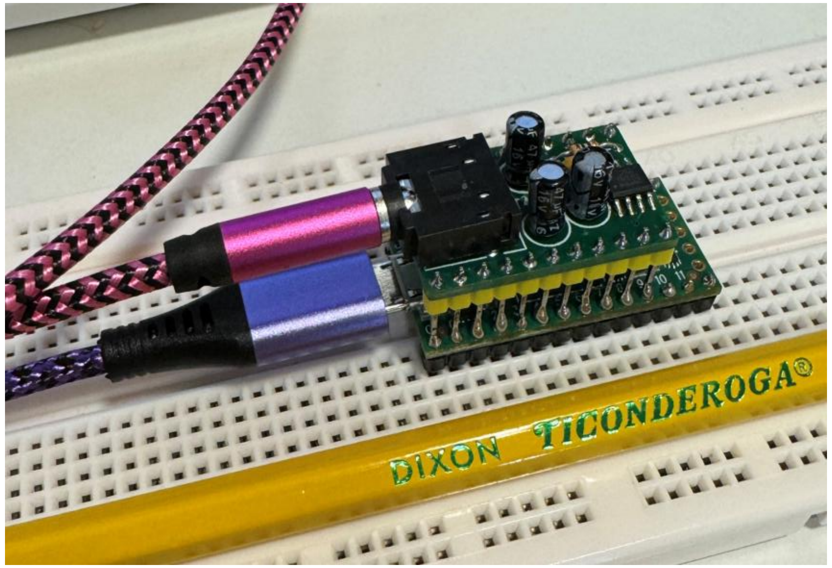
Code emulation

- Using SIMH NOVA -> Started running Network IV code
- Timer interrupts working out of the box!
- Modify SIMH NOVA to handle Device 76 (octal) IO
- Initial print-based debug showed signs of “life”
- Move SIMH to Raspberry Pi
- Redirect to IO to sound/input and display Teensy devices
- The original code is running with no changes from the converted paper tape

Sound

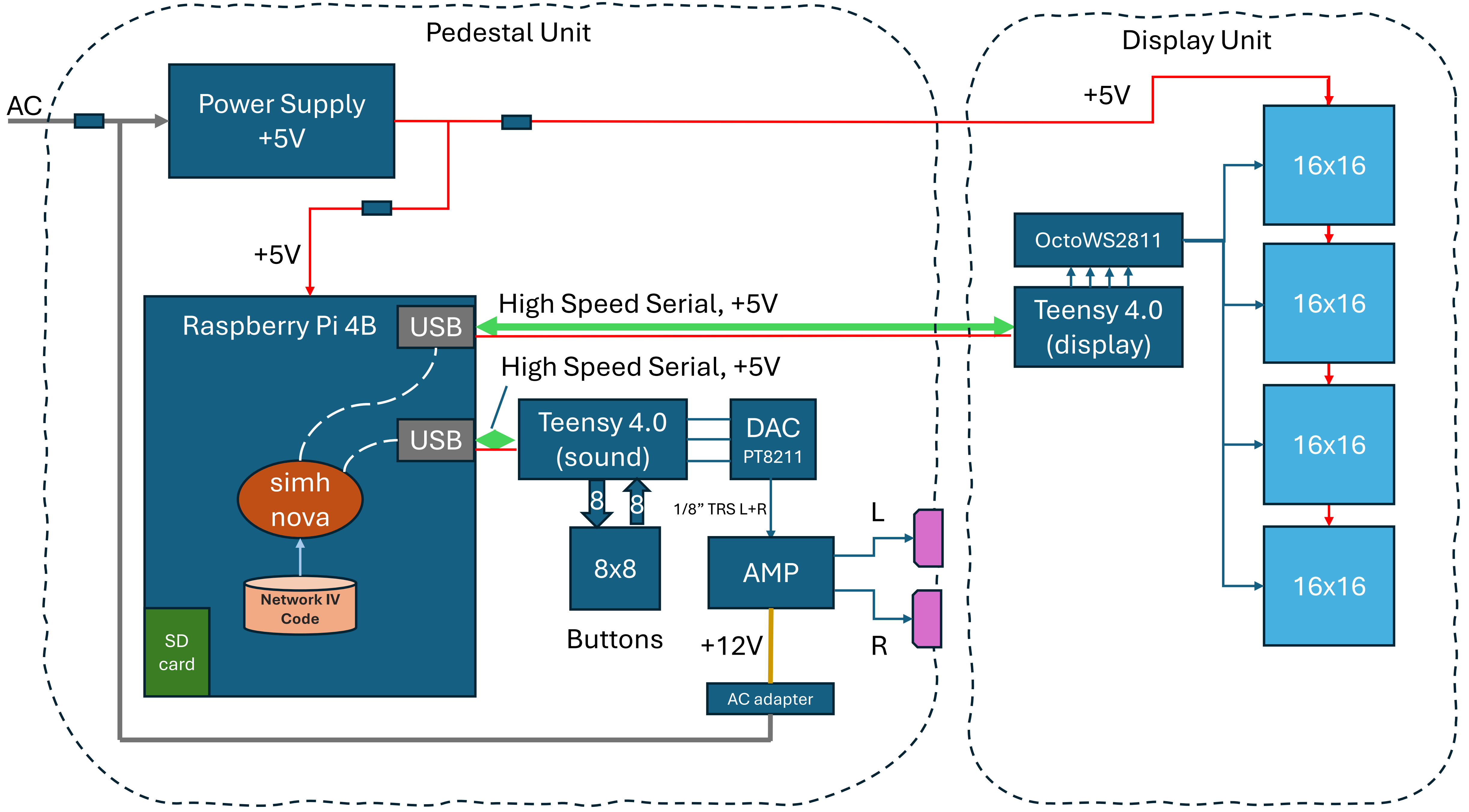
- Emulate sound using a Teeny 4.0 + DAC PT8211
  - 600MHz ARM w floating point
- Compute the analog waveforms digitally
- Started with resonators
  - Trio of decaying sinusoids
  - 32 resonators
- Added the three instruments
  - Oscillators
  - VCA
  - Ring modulator
  - Voltage controlled filters are not modeled yet
- Sounds controlled via simple fast serial commands over USB

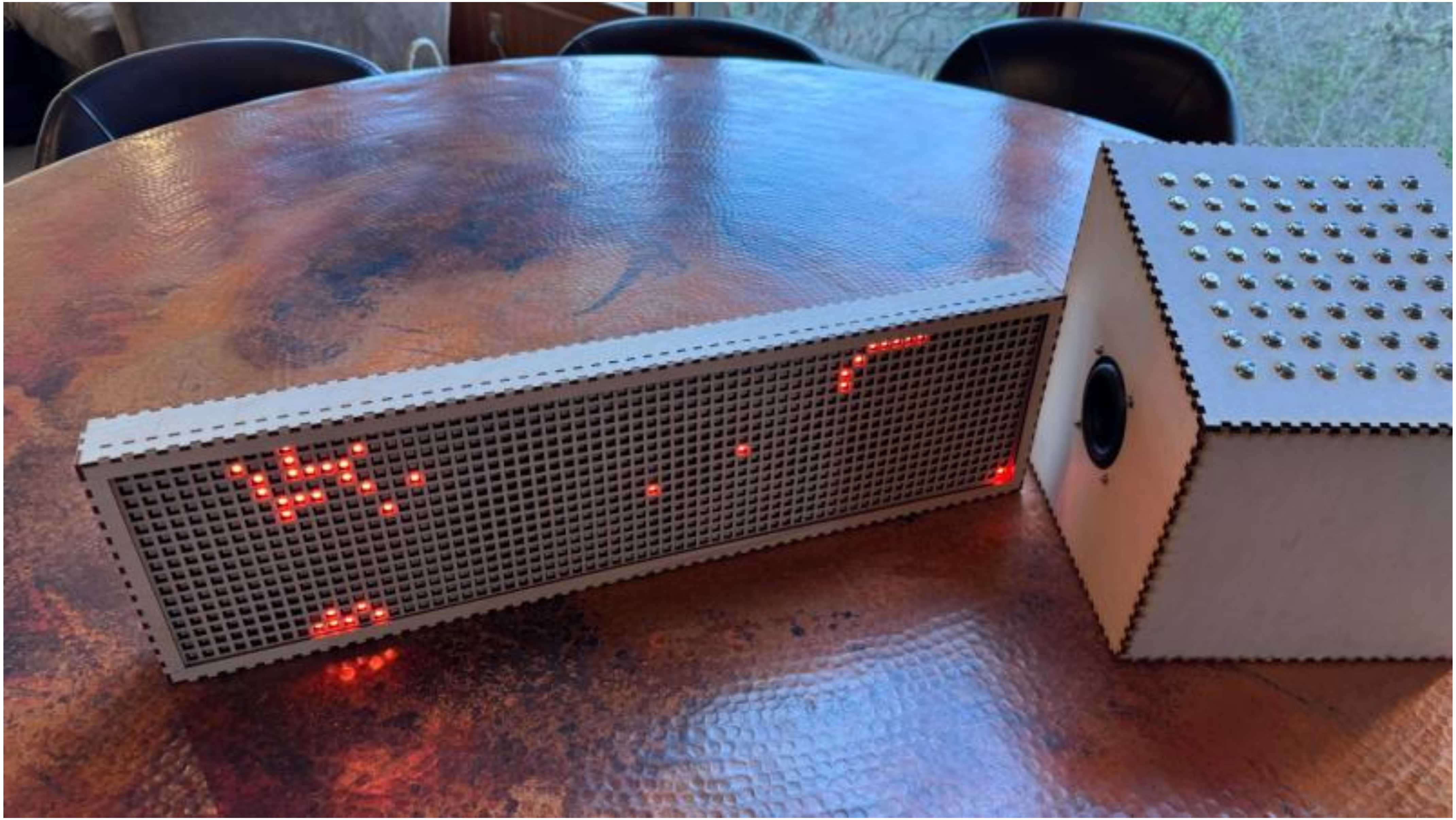
  r 5
  s 1 101
- Sound controller also reads button status



Display

- Use another Teensy for handling the display
- Four 16x16 addressable LED matrix displays
- Control with serial commands
  - Clear display (c)
  - Light a particular light (l <r> <c>)
  - Update a whole row of 64 lights with 16 hex chars (r <r> <hexdata>)


Pedestal Unit
Display Unit
AC
Power Supply
+5V
+5V
+5V
Raspberry Pi 4B
USB
USB
High Speed Serial, +5V
High Speed Serial, +5V
simh
nova
Network IV
Code
SD
card
Teensy 4.0
(sound)
DAC
PT8211
8
8
8x8
Buttons
1/8" TRS L+R
AMP
L
R
+12V
AC adapter
OctoWS2811
Teensy 4.0
(display)
16x16
16x16
16x16
16x16



Two Network IV Rebooted!

Future Directions

- Add MIDI capability
  - Add configurability to redirect certain instruments / sound parameters to certain MIDI channels
  - Mix Teensy sound emulation with synthesizers including real analog synths
- Larger installation?
- Revive using real a Nova system?

Acknowledgements

- Charles Keagle
- SIMH Project

Links

- James Seawright
  - https://www.seawright.net/
- Network IV
  - https://www.seawright.net/sculpture/opus20
- Network IV rebooted
  - https://www.seawright.net/network-iv-rebooted
- GitHub
  - https://github.com/NRGN2ART/Network-IV
  - https://github.com/NRGN2ART/PaperTapeReader
- Youtube
  - Network IV Rebooted